Introduction to Computer Programming
Fall 2026
Midterm

First Name	
Last Name	
NetID	

Instructions

- **Duration**: 9:50 AM-11:30 AM (100 minutes)
- **Max score**: 100 pts

- Computers, calculators, phones, textbooks or notebooks are **not allowed**
- When writing programs, make very clear your line **indentations** (there are some indentation levels in the areas where you will write code). If indentation is not perfectly clear, the grader will consider "no indent".
- We can **not answer** questions during the test.
- If you are unsure of a problem, please write down your assumptions, and proceed with it. We will grade accordingly.
- **Do not** use lists, string methods (such as .split() or .count()), dictionary, set or define functions. Use only concepts covered in class.

Grading

Questions	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**
Points	$8'$	$8'$	$24'$	$10'$	$10'$	$10'$	$15'$	$15'$
Score								
Total								

Question 1 | Boolean Expression

Evaluate the following boolean expression:

`'ace' in 'space' or 5%2 < 1`	
`3 == '3' or 'pen' > 'pencil'`	
`7//3 > 2 or len('abs') == 3 or not '12' == 12`	
`not True and True and True`	

Question 2 | Binary and Hexadecimal Representaion

Complete the table below by converting each given number into the other two number systems: decimal, binary, and hexadecimal.

Decimal	Binary	Hexadecimal
	`0b101011`	
		`0x5A`

Question 3 | Predict the Output

Program 1

```
print( 21 // 5 )
print( 21 % 5 )
print( f'{21/5:.2f}' )
```

Output 1

Program 2

```
a = 2
b = 5
print(a , b, sep='*', end = '=')
print(a * b)
print('Done!')
```

Output 2

Program 3

```
word = "PROGRAM"

print(word[3])
print(word[-4:len(word)]
print(word[:3])
print(word[1:-1:2])
print(word[-2::-2])
```

Output 3

Program 4	Output 4

```
count = 0

for i in range(1, 13):
    if count > 4:
        break

    if i % 2 == 0 or i % 3 == 0:
        print(i)
        count += 1

print("End")
```

Program 5	Output 5

```
res, n = 1, 1

while n < 5:
    res *= n
    n += 1

print(res)
```

Program 6	Output 6

```
result = ''

for ch in 'bAeBe3':
    if 'a' < ch < 'z':
        result += ch

print(result)
```

Program 7	Output 7

```
S = '1234'

if len(S) > 3:
    print('One')
elif int(S) % 2 == 0:
    print("Two")
else:
    print("Three")
```

Program 8	Output 8

```
total = 0

for i in range(7, 1, -2):
    total += i

print(total)
```

Question 4 | Predict the Output

```
N = aString

i, j = 0, 0
a = 0

while i < len(N):

    j = i + 1
    while j < len(N) and N[j] == N[i]:
        j = j + 1

    if j != i + 1:
        a = a + 1

    i = j

print(a)
```

Fill in the table with the expected output for each value of `aString`.

`aString`	Expected output
`'abc'`	
`'aa'`	
`'aabbcc'`	
`'aaaabccdeeffffgg'`	

Question 5 | Character Classifier

Write a Python program that reads a single character and classifies it as follows:

- If it is an English letter:
  - Display whether it is ***uppercase*** or ***lowercase***.
  - Display whether it is a ***vowel*** `(a,e,i,o and u)` or a ***consonant***.
- If it is a digit from `0` through `9`, display whether it is ***even*** or ***odd***.
- Otherwise, print ***Other character***.

Hint: *Use 'A' <= letter <= 'Z' to check whether letter is an uppercase letters; similarly for lowercase letters and digits*

Example Output (user input is in bold)**:**

Example 1:

```
Enter a character: E
Uppercase vowel
```

Example 2:

```
Enter a character: b
Lowercase consonant
```

Example 3:

```
Enter a character: 7
Odd digit
```

Example 4:

```
Enter a character: #
Other character
```

Write your code on the next page!

Your code:

Question 6 | Saving Goal

Write a program that asks for an initial deposit, an annual interest rate (in %), and a target amount. Each year the balance grows by the interest rate.

The balance of next year can be calculated as:

$$balance * (1 + rate/100)$$

Print the balance at the end of each year (2 decimal places) until the balance reaches or exceeds the target, then print how many years were needed.

Example Output (user input is in bold)**:**

Example 1:

Initial deposit: **1000**
Annual interest rate (%): **4**
Target amount: **1200**
Year 1: 1040.00
Year 2: 1081.60
Year 3: 1124.86
Year 4: 1169.86
Year 5: 1216.65
Years needed: 5

Write you code on the next page!

Your code:

Question 7 | Vending Machine

A vending machine only accepts `1, 5, 10` and `20` yuan bills/coins. Write a Python program that:

- asks for the item price (you may assume it is a positive integer, in yuan);
- repeatedly asks the user to insert money until the total paid is at least the price. Reject any value that is not `1, 5, 10` or `20` by printing ***Rejected: <value>***. After each accepted payment, print the progress;
- returns the change using **as few `10, 5` and `1` yuan pieces as possible**, printing only the denominations that are actually used. If there is no change, print ***No change. Enjoy your drink!***

Example Output (user input is in bold)**:**

Example 1:

Item price (yuan): **7**
Insert money (1, 5, 10 or 20): **3**
Rejected: 3
Insert money (1, 5, 10 or 20): **20**
Paid: 20 / 7
Change: 13
10 yuan x 1
1 yuan x 3
Enjoy your drink!

Example 2:

Item price (yuan): **8**
Insert money (1, 5, 10 or 20): **5**
Paid: 5 / 8
Insert money (1, 5, 10 or 20): **2**
Rejected: 2
Insert money (1, 5, 10 or 20): **1**
Paid: 6 / 8
Insert money (1, 5, 10 or 20): **1**
Paid: 7 / 8
Insert money (1, 5, 10 or 20): **1**
Paid: 8 / 8
No change. Enjoy your drink!

Write you code on the next page!

Your code:

Question 8 | Prime Finder

Write a Python program that reads two integers **`start`** and **`end`** (you may assume `start ≤ end`) and prints all prime numbers in that range (inclusive) on one line, separated by spaces, followed by the total count.

A prime is an integer greater than `1` whose only divisors are `1` and itself.

Example Output (user input is in bold)**:**

Exampe 1:

Start: **10**
End: **30**
11 13 17 19 23 29
Total primes: 6

Example 2:

Start: **1**
End: **10**
2 3 5 7
Total primes: 4

Your code:

ICP Cheat Sheet

Binary, Hexadecimal

```
bin(10)      0b1010
hex(90)      0x5A
```

Data type conversion

```
float('2')   2.0     float(2)     2.0
int('1')     1       int(1.9)     1
str(12)      '12'    str(12.1)    '12.1'
```

Scientific notation

```
10e2  #1000.0        10e-2   #0.1
```

Escape sequence

\n	new line	\\	back slash
\t	tab (8 characters)	\'	single quote

Data type check

```
type(12)     int     type('ICP')   str
type(1.0)    float   type(True)    bool
                     type(False)   bool
```

Comparison

```
1 == 1      True    1 != 1       False
1 < 2       True    2 <= 2       True
1 > 2       False   2 >= 2       True
1=='1'      False   1 ! = ' 1 '   Ture
'a'>'abc'   False   'abs'>'abd'  True
```

Membership

```
'a' in 'abc'       #True
'a' not in 'abc'   #False
'a' in 'ICP'       #False
'a' not in 'ICP'   #True
```

Boolean operators

```
not True        #False      not False #True
True and True   #True
True and False #False
False and False #False
True or True #True          True or False #True
False or False #False
```

Division

```
5/2       # 2.5
```

division operation always returns a float

Integer division

```
5//2      # 2          -5//2     #-3
6//4.0    # 1.0
```

Modulo

```
5%2       # 1          -5%2      # 1
10%4.0    # 2
```

Operator precedence (For highest to lowest)

()	parentheses
**	exponentiation
+, -	unary + and - (signs)
*, /, //, %	multiplication, division integer division and modulo
+,-	addition and subtraction
==, !=, <, <=, >=, >, in, not in	comparison operators/membership
not and or	Logic operators

Print function

```
print('a','b',sep='*')            # a*b
print('c','d',sep='?',end='*')    # c?d*e f
print('e','f')
```

Input function

```
name = input("Enter your name: ")
print("Your name is: " + name)
#Enter your name: ICP
#Your name is: ICP
```

range function

```
range(2,10,2)        range(10,0,-2)
#2,4,6,8             #10,8,6,4,2
range(1,4)           range(4,1)  #empty
#1,2,3               sequence
range(4)             range(-4)
#0,1,2,3             #empty sequence
```

Module

```
import math, random
math.sqrt(4)   #2       math.ceil(1.2) #2
math.floor(1.2)#1

random.random()
#a random float between 0 and 1(included)
random.randint(1,10)
#a random int between 1 and 10(included)
```

Decision Structure

```
if n == 0:
   print("n is zero")
elif n > 0:
   print("n is strictly positive")
else: # n < 0
   print("n is strictly negative")
```

Repetition Structure

```
n=0                          0
while n < 4:                 1
  print(n)                   2
  n+=1                       3
```

```
num=input('A number(0 to quit): '))
while num != 0:
    num=input('A number(0 to quit): '))
```

```
for i in range(3):           0
    print(i)                 1
                             2
```

```
for i in 'abc':              a
    print(i)                 b
                             c
```

String Delimiters

Single quotes: `'I am a string'`
Double quotes: `"I'm a string"`
Triple single quotes: `'''I'm a string'''`
Triple double quotes: `"""I'm a string"""`

f-string

```
print(f'Cost is {12.3456:.2f}$!')
#Cost is 12.35$!
```

String Operations

Concatenation : `'ABC'+'3'` `#'ABC3'`
Repetition: `'ABC'*3` `#'ABCABCABC'`

String Length

```
len('0123456789') # 10
len('\n')  # 1    len('') # 0
```

Unicode Code/Text Conversion

```
ord('a') #97          chr(97)  # 'a'
ord('A') #65          chr(65)  # 'A'
chr(ord('A')+3) # 'D'
```

String Indexing `S='ABCDEF123456'`

```
len(S) # 12
S[0]   # 'A'        S[-1]  # '6'
S[11]  # '6'        S[-12] # 'A'
```

String Slicing `S='ABCD1234'`

```
S[:4]    #'ABCD'    S[-4:]   #'1234'
S[4:]    #'1234'    S[-6:-2] #'CD12'
S[2:6]   #'CD12'    S[-1:-8] #''
S[4:1]   # ''       S[:5:2]  #'AC1'
S[::-1] #'4321DCBA'
S[1:7:2]#'BD2'      S[7:0:-3]#'41B'
```